(a) Teacher inputs and outputs

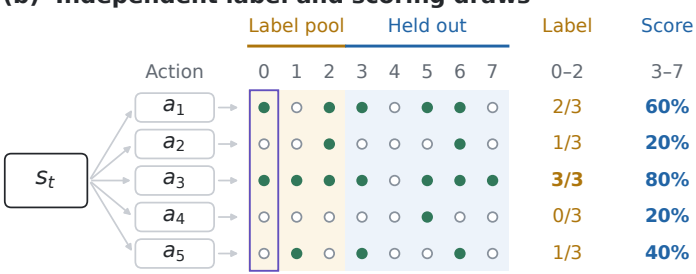
Shared input: task + history + current state s_t + all five candidate texts $a_1, \ldots, a_5$

Luna sees no outcome verdicts

Label source	Additional input for each candidate a_i	Operation	Output: choice	Score on 3–7
Pre-execution	Shared input only; no execution	GPT-6 Luna	a_1	60%
Post-execution	a_i → s_{t+1}	GPT-6 Luna	a_1	60%
Lookahead K = 3	a_i → s_{t+1} —u_1→ s_{t+2} —u_2→ s_{t+3} —u_3→ s_{t+4}	GPT-6 Luna	a_3	80%
Observed outcome	Draws 0–2: [2/3, 1/3, 3/3, 0/3, 1/3]	Argmax rate	a_3	80%

Draw 0 only; u_1, u_2, u_3: actor steps. K = 1 stops at s_{t+2}.

(b) Independent label and scoring draws



(c) Train a pre-action selection head

Pre / lookahead: vote shares $p_i = n_i/N$

Outcome: $p = (0, 0, 1, 0, 0)$ in this example

Train: 1,380 states · test: 348 states, 86 tasks

Uniform / SFT sample: 44% · oracle: 80%

Illustrative values · valid draws only. Label construction and head training never use draws 3–7.